Matrix Equations and Solutions Sets

Recall from last class:

- A matrix is in **echelon form** if the following 3 properties hold:
  1. All non-zero rows are above zero rows.
  2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.
- A matrix in echelon form is in **reduced echelon form** if
  1. The leading entries are all 1.
  2. Each leading 1 is the only non-zero entry in its column.
- Each matrix is row equivalent to one and only one reduced echelon matrix.
- A **pivot position** in a matrix A is a location in A that corresponds to a leading 1 in the reduced row echelon form. A **pivot column** is a column that has a pivot position.
- **Row Reduction Algorithm (Gaussian Elimination).**
  1. The leftmost non-zero column is a pivot column.
  2. Select a non-zero entry in the pivot column as pivot. Interchange rows to move this entry to the top.
  3. Use elementary row operations to create zeros below the pivot.
  4. Apply 1-3 to the submatrix obtained by removing pivot row and column. Repeat until all non-zero rows are modified.
  5. Beginning with rightmost pivot and working upwards and to the left, create 0's above each pivot. Scale pivot positions entries to be 1.
- The variables corresponding to the pivot columns in an augmented matrix are called **basic** and the others are called **free**.
- A linear system is **consistent** if and only if the rightmost column in the augmented matrix is not a pivot column. If the system is consistent, then it has a unique solution when there are no free variables and infinite solutions when there is one or more free variables.
- A matrix with 1 column is called a **column vector**. A matrix with one row is called a **row vector**.
- We can add vectors of the same size by adding the corresponding components; namely, if $\vec{u} = \begin{bmatrix} u_1 \\ \vdots \\ u_n \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} v_1 \\ \vdots \\ v_n \end{bmatrix}$, then $\vec{u} + \vec{v} = \begin{bmatrix} u_1 + v_1 \\ \vdots \\ u_n + v_n \end{bmatrix}$. Similarly, we can scale a vector by a real number c, by multiplying all the components by c; namely, $c\vec{u} = \begin{bmatrix} cu_1 \\ \vdots \\ cu_n \end{bmatrix}$.

- Given $\vec{v}_1, \dots, \vec{v}_p \in \mathbb{R}^n$ and scalars $c_1, \dots c_p$, the vector

$$\vec{y} = c_1\vec{v}_1 + \dots c_p\vec{v}_p$$

  is called the **linear combination** of $\vec{v}_1, \dots, \vec{v}_p$ with weights $c_1, \dots, c_p$.

- A vector equation

$$x_1\vec{a}_1 + x_2\vec{a}_2 + \cdots + x_n\vec{a}_n = \vec{b}$$

  has the same solutions as the augment matrix

$$[\vec{a}_1\ \vec{a}_2\ \dots\ \vec{a}_n \big| \vec{b}].$$

  Hence $\vec{b}$ is a linear combination of $\vec{a}_1, \dots, \vec{a}_n$ if and only if there is a solution to this matrix.

- If $\vec{v}_1, \dots, \vec{v}_p \in \mathbb{R}^n$, then the set of all linear combinations of $\vec{v}_1, \dots, \vec{v}_p$ is called the **span** of $\vec{v}_1, \dots, \vec{v}_p$. We write this set as $\text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$.

Let A be an $m \times n$ matrix, then

$$A = [\vec{a}_1 \ \dots \ \vec{a}_n], \quad \vec{a}_j \in \mathbb{R}^m, \ 1 \le j \le n.$$

If $x \in \mathbb{R}^n$, then we define

$$A\vec{x} = \vec{a}_1 x_1 + \vec{a}_2 x_2 + \cdots + \vec{a}_n x_n.$$

Thus the matrix equation $A\vec{x} = \vec{b}$ is the same as the vector equation

$$\vec{a}_1 x_1 + \vec{a}_2 x_2 + \cdots + \vec{a}_n x_n = \vec{b}.$$

1. Given

$$A = \begin{bmatrix} 1 & 3 & -4 \\ 1 & 5 & 2 \\ -3 & -7 & 6 \end{bmatrix} \text{ and } \vec{b} = \begin{bmatrix} -2 \\ 4 \\ 12 \end{bmatrix}.$$

Is $A\vec{x} = \vec{b}$ consistent?

Fact: $A\vec{x} = \vec{b}$ has a solution if and only if $\vec{b} \in \text{Span}\{\vec{a}_1, \ldots, \vec{a}_n\}$ where $\vec{a}_j$ is the j-th column of A.

2. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$. Is $A\vec{x} = \vec{b}$ consistent for all $\vec{b} \in \mathbb{R}^3$.

Definition 1. *A set of vectors $\vec{v}_1, \ldots, \vec{v}_p$ in $\mathbb{R}^m$* **spans** *$\mathbb{R}^m$ if every vector in $\mathbb{R}^m$ is a linear combination of $\vec{v}_1, \ldots, \vec{v}_p$.*

Theorem Let A be an $m \times n$ matrix, then the following are equivalent.

(a) For each $\vec{b} \in \mathbb{R}^m$, the matrix equation $A\vec{x} = \vec{b}$ is consistent.

(b) Each $\vec{b} \in \mathbb{R}^m$ is a linear combination of the columns of A.

(c) The columns of A span $\mathbb{R}^m$.

(d) A has a pivot position in every row.

3. Prove the theorem.

Fact (or Theorem 5 from the book): If A is a $m \times n$ matrix and $\vec{u}, \vec{v} \in \mathbb{R}^n$, $c \in \mathbb{R}$, then

(a) $A(\vec{u} + \vec{v}) = A\vec{u} + A\vec{v}$

(b) $A(c\vec{u}) = c(A\vec{u})$.

4. Prove the result.

5. **True or False**. If the augmented matrix has a pivot position in each row, then it is consistent.

6. Let

$$\vec{u} = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix} \text{ and } A = \begin{bmatrix} 2 & -5 \\ 2 & 6 \\ 1 & 1 \end{bmatrix}.$$

Is $\vec{u}$ in the span of the columns of A?